

Jacob Vaught

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EDUCATION

University of South Carolina, Columbia, SC GPA: 3.56
B.S. Electrical Engineering, B.S. Computer Science, Minor in Mechanical Engineering Aug 2024 – May 2028

Midlands Technical College, Columbia, SC GPA: 3.89
Certificate in Engineering Sciences Jan 2022 – May 2024

Honors: Dean's List (2x), President's List (2x), Magna Cum Laude, Palmetto Fellows Scholarship

RESEARCH EXPERIENCE

ARTS-Lab (Advanced Real Time Systems Laboratory) Columbia, SC
Undergraduate Research Student, Dr. Austin Downey Feb 2025 – Present

- Developed FEA models for vibration analysis of complex structures in structural health monitoring
- Implemented FPGA-based system to generate and read high frequency signals for real-time analysis
- Co-authored IMAC paper: Modal analysis using a UAV-deployable wireless sensor network
- Awarded NSF grant for continued research in Structural Health Monitoring

McNair Aerospace Center Columbia, SC
Undergraduate Research Student, Dr. Lang Yuan Aug 2024 – Present

- Developed FEA models for Laser Powder Bed Fusion Additive Manufacturing (LPBFAM) using ABAQUS
- Authored grant proposal for continued research into computational aspects of LPBFAM
- Awarded Magellan Scholar Grant (2x) and McNair Junior Fellow Grant for research contributions

PROFESSIONAL EXPERIENCE

Westinghouse Electric Company LLC Hopkins, SC
Summer Intern May – Aug 2025

- Designed and modeled component to prevent lateral movement of torque testing rig for TP-BAR
- Developed Python programs to analyze grid growth in 3 nuclear fuel assemblies (Vogtle Unit 2)
- Conducted preliminary analysis on potential switch from 18-month to 24-month fuel cycles
- Contributed to project increasing fuel enrichment from 5% to 6%, reducing costs for consumers

University of South Carolina Columbia, SC
Teaching Assistant, Thermodynamics & Mechatronics Aug 2025 – Present

- Graded work for 50+ students including exams within 48-hour turnaround
- Assisted students with complex concepts during Q&A sessions for Dr. Sepahvandi and Dr. Vitzilaios

LEADERSHIP & ORGANIZATIONS

IEEE	President
SAE	President
ASME	Secretary
SAMPE	Secretary
Student Senate	Senator, 115th Session

SKILLS

Programming:	Python, Java, MATLAB, C++, C, Assembly, JavaScript, HTML/CSS, Shell
CAD & Simulation:	SOLIDWORKS, AutoCAD, Autodesk Inventor, Creo, ABAQUS, SPICE, Logisim
Hardware:	Arduino, FPGA, Oscilloscope, Function Generator, Soldering